High-Performance Under Sparse Annotations: Cross-Prototype Bidirectional Constrained Semi-Supervised Framework for Remote Sensing Image Semantic Segmentation

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***Abstract*—Class imbalance has long posed a significant challenge to semantic segmentation in remote sensing imagery (RSI). In semi-supervised settings, existing studies attempt to alleviate this issue through data augmentation, loss function refinement, and optimized training strategies. However, these methods generally rely on an idealized assumption that labeled and unlabeled data share the same class distribution. In practice, unlabeled data typically far exceeds labeled data in scale, making this assumption difficult to satisfy. Such distributional inconsistency not only weakens the effectiveness of semi-supervised learning but also amplifies the inherent class imbalance in remote sensing segmentation. To address these key challenges, we propose a cross-prototype bidirectional constraint framework (CPBCF) for semi-supervised semantic segmentation (SSSS). The framework aligns unlabeled prototypes with labeled feature representations to enforce class consistency among unlabeled prototypes, preventing them from being overwhelmed by majority-class features in the early training stages and reducing pseudo-label bias. In addition, by imposing class-level constraints between unlabeled features and labeled prototypes, CPBCF enables the learning of minority-class features to be reinforced by information from unlabeled prototypes, thereby improving the model's ability to recognize rare categories. To further suppress majority-class dominance, a dynamic class-weight adjustment mechanism is introduced, which selectively applies prototype constraints to minority classes. Through this prototype-level bidirectional balancing strategy, CPBCF enhances the model's capacity to learn robust minority-class representations under the guidance of unlabeled data, effectively mitigating performance degradation caused by class imbalance in semi-supervised scenarios. Extensive experiments on the ISPRS Potsdam and Vaihingen datasets show that our method consistently surpasses state-of-the-art approaches, achieving notable improvements across all benchmarks.**

***Index Terms*—Category distribution, class imbalance, prototype learning, remote sensing images, semi-supervised semantic segmentation (SSSS).**

I. Introduction

SEMANTIC segmentation of remote sensing imagery (RSI) depends heavily on high-quality, large-scale labeled datasets [1], [2], [3], [4]. However, in real-world applications, label scarcity and class imbalance remain persistent challenges [5], [6]. Semi-supervised semantic segmentation (SSSS) alleviates the annotation burden by leveraging unlabeled data [7], [8], yet most existing studies have overlooked the representation bias induced by class imbalance [9]. Due to limited labeled samples and skewed class distributions, models often fail to learn discriminative features for rare classes, thereby further amplifying the imbalance issue in semi-supervised learning.

Class imbalance has long been a major obstacle to advancing semantic segmentation models and is particularly pronounced in RSI. Its underlying causes can be attributed mainly to two factors. First, the natural scarcity of certain classes: In natural image datasets, the proportion of rare classes can often be adjusted through targeted data collection or annotation strategies. **In contrast, in RSI, the wide observation range, complex scene composition, and limited human controllability lead to inherently imbalanced land-cover distributions, with some rare classes appearing only in specific regions or scenes. Second, scale disparity: Different land-cover categories display substantial variations in spatial scale, and remote sensing, as a long-range observation modality, further amplifies these differences, thereby intensifying the imbalance problem.** For example, in the ISPRS Potsdam dataset, the car class not only contains far fewer samples but also occupies a much smaller spatial extent than the building class, resulting in an extremely low pixel proportion overall. Consequently, the model struggles to learn representative features for such rare categories [as illustrated in Fig. 1(a)]. To alleviate class imbalance, previous

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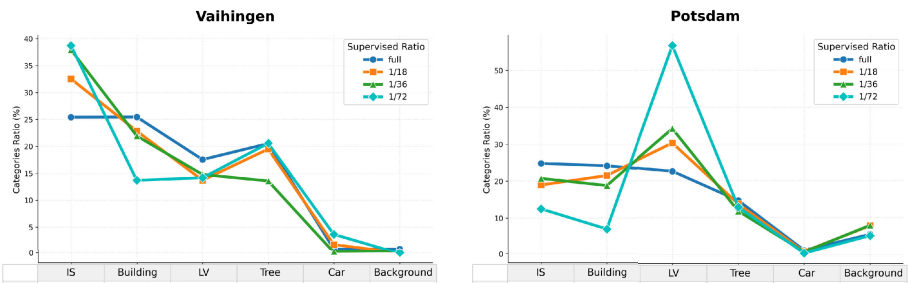
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(a) Scale disparity



Vaihingen	IS	Building	LV	Tree	Car	Background
full	25.43	25.46	17.54	20.53	0.89	0.89
1/18	32.55	22.85	13.7	19.51	1.72	0.18
1/36	37.96	21.93	14.76	13.56	0.26	0.36
1/72	38.73	13.68	14.16	20.58	3.63	0.05

Potsdam	IS	Building	LV	Tree	Car	Background
full	24.83	24.18	22.62	14.67	1.15	5.48
1/18	18.98	21.53	30.29	13.74	0.79	7.83
1/36	20.76	18.82	34.19	11.79	0.72	7.92
1/72	12.51	7.04	56.74	12.88	0.37	5.12

(b) Ratio of different categories under various supervised ratios

Fig. 1. Illustration of class imbalance. (a) Object size differences among various classes in remote sensing images. (b) Class proportion variations under different labeled data settings within the Vaihingen and Potsdam datasets.

studies have primarily explored data-level, architectural-level, and loss-level [10] optimization strategies.

However, most of these approaches assume an adequate number of labeled samples and a stable class distribution. In SSSS of RSI, the scarcity of labeled data and the presence of pseudo-label noise make these methods difficult to apply effectively. Moreover, these limitations often cause models to overfit majority classes, thereby further exacerbating the imbalance problem.

A detailed analysis shows that the class imbalance problem in SSSS is more complex and exhibits a coupling amplification effect. Existing methods typically rely on labeled data for explicit supervision while leveraging unlabeled data through high-confidence pseudo-labeling strategies. However, due to the inherent characteristics of RSI—namely large intraclass variability and small interclass differences, the model struggles to learn stable and discriminative representations when labeled samples of certain classes are extremely scarce. Although the large-scale unlabeled data may contain abundant minority-class samples that could theoretically alleviate class scarcity, the absence of explicit supervision prevents the model from learning meaningful features for these classes during early training. **As a result, minority-class samples in the unlabeled set are frequently misclassified or assigned low-quality pseudo-labels, allowing majority-class features to dominate the learning process. Consequently, valuable information within the unlabeled data remains underutilized, and the class imbalance problem persists or is even further amplified.** Specifically, Fig. 1(b) and (c) illustrates this issue using the ISPRS Vaihingen and Potsdam datasets. The blue curve shows the class distribution over the full training set. Under sparse sampling, the class distribution of the labeled subset often deviates significantly from the global distribution. For instance, "Buildings," a majority class in the complete dataset, may become a rare class in a 1/36 labeled subset. This distributional inconsistency forces the model to learn from biased priors, which is a key challenge addressed by our proposed framework.

In summary, the model's weak ability to recognize rare classes leads to unreliable pseudo-labels for samples, hindering the full utilization of potentially informative. As a result, the class imbalance problem remains insufficiently addressed. Therefore, effectively mitigating pseudo-label bias caused by class imbalance and enhancing the recognizability of rare classes under limited annotation conditions have become critical challenges restricting the further development of SSSS in RSI.

To tackle these challenges, we propose a cross-prototype bidirectional constraint framework (CPBCF) for SSSS of RSI. By modeling the distribution of category semantic prototypes, we construct prototypes for both labeled and unlabeled data and introduce a bidirectional constraint mechanism to dynamically balance the learning intensity across categories. By aligning unlabeled prototypes with labeled features, the framework enforces class consistency on unlabeled prototypes, preventing them from being dominated by majority-class features during early training and mitigating pseudo-label bias. Conversely, through class-level constraints between unlabeled features and labeled prototypes, the knowledge captured by labeled prototypes guides the learning of minority-class representations, enhancing the model's ability to recognize rare classes. Moreover, to better capture the true distribution of the training data, the framework dynamically tracks class frequencies and incorporates an adaptive class-weight adjustment strategy that updates weights based on real-time class distributions, thereby strengthening the learning of minority classes and suppressing the dominance of majority classes. Through this prototype-level bidirectional balancing mechanism, CPBCF substantially improves the model's ability to learn minority-class representations under the guidance of unlabeled data, effectively alleviating performance degradation caused by class imbalance in semi-supervised settings. Specifically, the main contributions of this article are as follows.

1) We propose a CPBCF. It models the category learning process in SSSS from the perspective of semantic prototype distributions and enables dynamic balancing of learning intensity across different categories.
2) We design a prototype-level bidirectional constraint mechanism that simultaneously maintains class consistency of unlabeled prototypes and enhances the model's capability to recognize rare categories.

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3) We develop an adaptive class-weight adjustment strategy to mitigate the adverse effects of class imbalance on model performance. This strategy updates class weights based on class distribution statistics during training.

II. Related Works

A. Class Imbalance in Supervised Learning

Current mainstream methods in semantic segmentation that address class imbalance generally optimize at the data level, loss function level, and model structure level.

At the data level, the core idea is to mitigate class imbalance by adjusting the distribution of training samples. These methods are typically applied during the data preprocessing stage and are independent of specific model architectures, providing strong generality and flexibility [11], [12]. Common strategies include oversampling minority-class samples [13], [14] or undersampling majority-class samples [15], [16]. However, oversampling may introduce noise or artificial patterns, whereas undersampling can lead to the loss of important information [17].

At the loss function level, improving the objective function to directly address class imbalance is another important research direction. Early approaches primarily employed class-weighting (CW) strategies, such as class-weighted cross entropy, to assign higher loss weights to classes with fewer pixels [18]. However, static weighting cannot account for variations in sample difficulty. Focal loss [10] guides the model to focus on hard-to-classify pixels by reducing the loss contribution of easy examples, and has proven effective in segmenting small objects and rare classes. Dice loss [19] and its variants—such as generalized Dice loss [20]—directly optimize the overlap between predicted and ground-truth regions, effectively mitigating gradient bias and achieving strong performance in highly imbalanced scenarios, particularly in medical image segmentation. Boundary loss [21] introduces a distance transform map to explicitly constrain the spatial discrepancy between predicted and true boundaries, improving the segmentation of fine-grained structures. It is often combined with Dice loss to leverage complementary strengths [22].

At the model-architecture level, recent research has increasingly focused on decoder design, structural decoupling, and attention mechanisms as key optimization strategies to enhance minority-class feature representation and improve the robustness of model predictions [23]. Multiexpert or multidecoder architectures [24], [25] mitigate the tendency of a single decoder to dominate tail-class features by employing independent branches for different categories or frequency components and integrating them at the output stage. Representation–classifier decoupling methods [26], [27] improve tail-class representations by separating the gradient flows of the encoder and decoder, thereby preventing the encoder from becoming overly biased toward head-class features. In addition, spatial and channel attention mechanisms [28], [29], [30] strengthen the model's ability to focus on minority-class pixels; multiscale or pyramid architectures [31] enhance the multiscale representation of small-object features; and boundary-aware modules [32], [33], [34] improve the model's capability to capture the edges and shapes of rare classes. These structural enhancements are typically embedded directly into the network architecture, making the model inherently more sensitive to minority-class information and thus more effective in handling class-imbalanced semantic segmentation tasks.

B. Class Imbalance in Semi-Supervised Learning

Although some ideas for handling imbalance in supervised learning (such as sample resampling and feature constraint) are still inspiring in SSSS, due to the existence of unlabeled data and the pseudo-label mechanism, semi-supervised learning shows more complex class imbalance characteristics. Existing studies mainly focus on improving the quality of pseudo-labels and modeling the class distribution to indirectly alleviate the performance degradation caused by class imbalance. Commonly used data augmentation strategies (e.g., Cutout [35], CutMix [36], and ClassMix [37]) can enhance the diversity and stability of pseudo-labels.

Similar to supervised learning, improving the loss function is also a key approach for SSSS to tackle annotation noise and class imbalance. Wang et al. [38] proposed class-aware cross entropy (CCE). This loss function introduces image-level label constraints, which helps reduce class errors in pseudo-labels. Xie et al. [39] designed a balanced loss function. It is used to weaken the dominance of common categories. Liu et al. [40] significantly enhanced training stability and accuracy. They achieved this by adding an auxiliary teacher network and replacing the original mean squared error with confidence-weighted cross entropy.

Besides the loss function aspect, optimizing training strategies has become another direction to improve model robustness. Huang et al. [41] proposed a decoupled weighted learning framework. It is used to suppress pseudo-label noise of tail classes. Wang et al. [42] constructed an unbiased subclass regularization network (USRN). They alleviated class imbalance by balancing the subclass distribution. Sohn et al. [43] first adopted a pseudo-label generation mechanism based on confidence filtering in semi-supervised segmentation. This work inspired a series of subsequent adaptive methods. Wang et al. [44] further proposed adaptive class fairness regularization. It dynamically adjusts the threshold to improve the utilization of tail classes. Chen et al. [45] used a truncated Gaussian weighting mechanism and a unified alignment strategy. These methods enhanced the mining efficiency of weak class samples. Zhu et al. [46] maintained memory banks for labeled and unlabeled data through class-ranked weighting. They dynamically calculated class weights based on the relative ranking of pixel confidence and adaptively fused these weights. Jin et al. [47] adopted dynamic pseudo-label sampling to mitigate class distribution imbalance. This is achieved by prioritizing classes with fewer samples.

III. Methods

A. Network Architecture

Before introducing the overall framework, we first describe the composition of the datasets used in this article. Suppose

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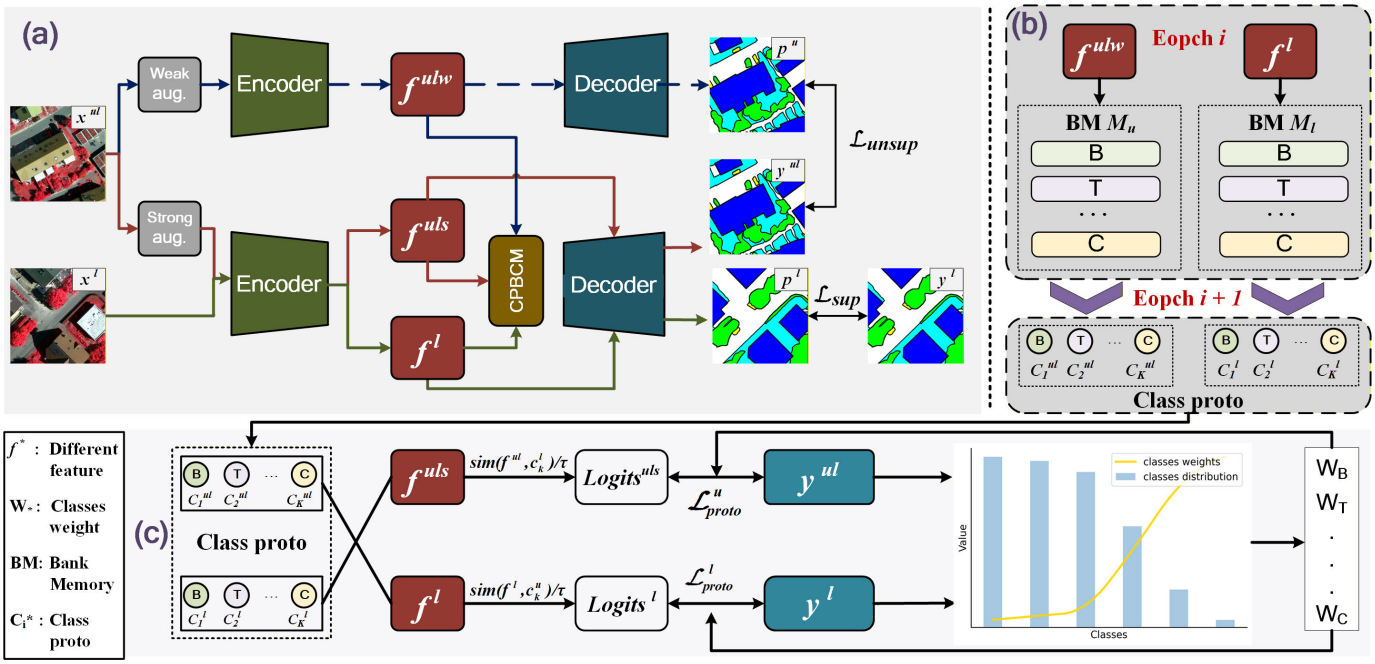


Fig. 2. Overall network architecture diagram of this article. (a) Flow directions of different data, where arrows of different colors represent the data flows of various data types, and dashed lines indicate no participation in backpropagation. (b) Epoch-wise zero momentum, where "Epoch i" denotes the ith epoch and "Epoch $i+1$" denotes the next epoch. (c) Bidirectional constraint mechanism.

there is a small labeled dataset D_l and a larger unlabeled dataset D_u, where $|D_l| \ll |D_u|$. To better improve the model's generalization ability, we apply strong augmentation x^{uls} (such as color transformation) and weak augmentation x^{ulw} (such as rotation and scaling) to the unlabeled data. The data flow of the weakly augmented unlabeled data does not participate in backpropagation; instead, it is used to provide reliable pseudo-labels for the strongly augmented data.

The overall framework is built on DeepLabV3 + [48], and the entire process is shown in Fig. 2. It consists of three supervision signals: the supervision loss of labeled data, the consistency loss of unlabeled data, and the bidirectional prototype supervision loss.

During the training of each epoch i, deep semantic features of the unlabeled data (i.e., f^{ulw} and labeled data f^l) are extracted and stored in the corresponding bank memory by category. After each epoch, category prototypes for each class are generated using the bank memory as shown in Fig. 2(b)], with detailed descriptions in Section III-B.

Meanwhile, during the training of each epoch, the category distribution is counted using the generated pseudo-labels, based on which category weights are generated (detailed in Section III-C). Using the generated semantic category prototypes and category weights, bidirectional prototype constraint learning is performed in the next epoch [as shown in Fig. 2(c)], with detailed explanations in Section III-D.

B. Update of Category Semantic Prototypes

As compact descriptors of category-wise feature distributions, category prototypes effectively aggregate the shared characteristics of samples belonging to the same class, thereby providing crucial support for addressing class imbalance. In RSI semantic segmentation, majority classes often possess abundant samples, causing their feature distributions to dominate the model's learning process. Conversely, minority classes typically suffer from insufficient feature representation and limited discriminability due to their scarce sample quantities.

The traditional prototype update mechanism based on exponential moving average (EMA) exhibits notable limitations in class-imbalanced scenarios. Because EMA incorporates accumulated historical information, minority-class prototypes can become overly influenced by past features of majority classes, resulting in feature drift. Specifically, even when the model gradually improves its ability to learn minority-class features, EMA-based prototypes remain affected by earlier low-quality representations and fail to promptly align with the latest feature distributions. This lag is particularly pronounced in RSI, where some minority classes (e.g., vehicles and small vegetation patches) account for less than 1% of total pixels. Their feature updates are easily overshadowed by fluctuations in the dominant classes (e.g., buildings and bare ground).

To overcome this limitation, we propose an Epoch-wise Zero Momentum strategy for re-estimating semantic category prototypes. The core idea is to eliminate cumulative dependencies on historical features and enforce strict alignment between prototypes and the current model state. The procedure is as follows.

At the beginning of each training epoch, prototype statistics (i.e., feature summations and pixel counts) are globally reset to zero, ensuring that no residual information from the previous epoch is retained. Throughout the epoch, features and labels (including pseudo-labels) from both the labeled and unlabeled branches are accumulated pixel-wise for each category. For labeled data, manually annotated ground-truth labels are used

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directly

$$M_l = \left\{ f_i^l \cdot \mathcal{I}\left(y^l = i\right) \right\}_{i=[1,K]} \tag{1}$$

where M_l denotes the set of valid feature-label pairs for labeled data, $\mathcal{I}(\cdot)$ is an indicator function that takes 1 when $y^l = i$ and 0 otherwise, y^l is the real label of the data, and K is the total number of categories.

For unlabeled data, only features corresponding to high-confidence pseudo-labels that meet the confidence threshold [as shown in (2)] are accumulated to filter out interference from low-quality pseudo-labels

$$M_u = \left\{ f^{\text{ulw}} \cdot \mathcal{I}\left(y^{\text{ul}} = i\right) \,\middle|\, \max\left(p_u\left(y^{\text{ul}}\right)\right) \geq \tau_p \right\}_{i=[1,K]} \tag{2}$$

where M_u is the set of valid feature-label pairs for unlabeled data, y^{ul} is the pseudo-label generated by the model for unlabeled data, p_u is the model's predicted probability distribution for unlabeled samples, and τ_p is the confidence threshold.

At the end of each training epoch, accurate prototype estimation is achieved through category-conditioned feature aggregation and statistical normalization. In the labeled data branch, features of all valid pixels are accumulated at the category level under the guidance of ground-truth labels. In the unlabeled data branch, only features associated with high-confidence pseudo-labels are aggregated to ensure reliability. Finally, the prototype vectors c^l and c^u for the current epoch are calculated using the following equations:

$$c_k^l = \frac{\sum_{(f_l, y_l) \in M_l,\, y_l = k} f_l}{N_k^l}, \quad c_k^u = \frac{\sum_{(f_u, \hat{y_u}) \in M_u,\, \hat{y_u} = k} f_u}{N_k^u} \tag{3}$$

where $c^l = \{c_k^l\}_{k=[1,K]}$ and $c^u = \{c_k^u\}_{k=[1,K]}$ are the prototype vectors of the kth category in labeled and unlabeled data, respectively, and N_k^l and N_k^u are the numbers of valid pixels for the corresponding categories.

This strategy ensures that category prototypes accurately track the model's current feature extraction capability in real time and remain sensitive to minority-class features that gradually emerge during training. Even under extremely low labeling rates, where category proportions may fluctuate drastically, the method can still robustly capture the feature distribution of each class in every epoch.

C. Category Distribution Statistics and Weight Update

As discussed previously, the distribution mismatch between labeled and unlabeled data may cause the model to develop biased judgments toward minority classes. Therefore, it is necessary to continuously update the model's assessment of minority classes as training progresses.

To determine minority classes, it is essential to accurately estimate the class distribution in the training data, considering both labeled and unlabeled data simultaneously. For labeled data, we traverse the training set to count the number of pixels for each class. For unlabeled data, we filter out high-reliability pseudo-labeled pixels that meet the confidence threshold and only accumulate the valid pixel counts for each class from this portion. Let n_k denote the valid pixel count for class k. The final class distribution is calculated as follows:

$$\hat{P}_k = \frac{n_k + \epsilon}{\sum_{k'=1}^{K} (n_{k'} + \epsilon)} \tag{4}$$

where n_k is the total number of pixels in the kth class and ϵ is a smoothing factor (set to 0.01 in experiments) to avoid numerical anomalies caused by zero pixel count for a class.

Based on the estimated category distribution, minority classes are determined. In this article, a minority class is defined as one whose pixel proportion is significantly lower than the average expected proportion. For the M categories involved in the adjustment (excluding boundary classes), the average expected proportion is $\bar{p} = 1/M$. A binomial test is employed to determine whether the pixel proportion of each class is significantly below the theoretical expectation. The null hypothesis states that the true pixel proportion of class c equals the expected value $\bar{p}$," while the alternative hypothesis states that it is significantly lower than $\bar{p}$." The corresponding test statistic is computed as follows:

$$p_value = \text{binomtest}\left(n_k, N, \bar{p}, \text{alternative} = \text{'less'}\right) \tag{5}$$

where N denotes the total number of pixels included in the statistical analysis and binomtest represents an exact hypothesis-testing function based on the binomial distribution. When $p_value < 0.05$, the null hypothesis is rejected, and class c is identified as a minority class. This statistically grounded mechanism eliminates bias arising from manually selected thresholds, thereby ensuring objective and stable minority-class identification.

The base weight is defined as inversely proportional to the estimated class frequency, $(w_k^{\text{base}} = 1/(\hat{P}_k + 1))$, which assigns higher initial weights to low-frequency classes. In addition, the base weights are normalized to have a mean of 1, preventing unintended scaling of the overall loss magnitude. For identified minority classes, their weights are amplified by an enhancement factor δ

$$w_k^{\text{e}} = \begin{cases} w_k^{\text{base}} \cdot \delta, & \text{if } k \in \text{minority classes} \\ w_k^{\text{base}}, & \text{if } k \in \text{majority classes.} \end{cases} \tag{6}$$

To prevent extreme weight values from affecting training stability, upper and lower bounds are set

$$w_k^{\text{clamped}} = \begin{cases} \min\left(w_k^{\text{e}}, w_{\max}\right), & \text{if } k \in \text{minority classes} \\ \max\left(w_k^{\text{e}}, w_{\min}\right), & \text{if } k \in \text{majority classes.} \end{cases} \tag{7}$$

This mechanism ensures that class weights can adapt to dynamic changes in data distribution during training, continuously optimizing the learning focus on minority classes.

D. Cross-Prototype Bidirectional Constraints Mechanism

To further enhance the discriminability of minority-class features, this article introduces a bidirectional prototype cross-constraint mechanism. This mechanism leverages the complementary strengths of labeled and unlabeled prototypes to construct bidirectional supervisory signals. The key idea is that labeled prototypes c_k^l provide reliable, strongly supervised semantic information, whereas unlabeled prototypes c_k^u capture the feature diversity inherent in large-scale data. By cross-referencing these two types of prototypes, their complementary advantages can be fully exploited. Specifically, the cross-constraint is implemented through two contrastive losses.

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1) Labeled Feature-Unlabeled Prototype Constraint: For a feature f_l of labeled data (with its real label $y_l = k$), it is required that its similarity to the unlabeled prototype c_k^u is higher than that to unlabeled prototypes of other classes. The loss function is defined as

$$\mathcal{L}_{\text{proto}}^{l} = \frac{1}{|M_l|} \sum_{(f_l, y_l) \in M_l} \mathcal{L}_{\text{CE}} \left(\frac{\text{sim}\left(f^l, c_k^u\right)}{\tau_t}, y_l \right) \tag{8}$$

where $\text{sim}(a, b)$ is the cosine similarity between feature vectors a and b, τ_t is the temperature coefficient, and $\mathcal{L}_{\text{CE}}$ is the cross-entropy loss. This loss forces labeled features to align with unlabeled prototypes, enhancing feature discriminability by utilizing information from more potential minority-class samples in unlabeled data.

2) Unlabeled Feature-Labeled Prototype Constraint: For a high-confidence unlabeled feature f_u (with its pseudo-label $\hat{y_u} = k$), it is required that its similarity to the labeled prototype c_k^l is optimal. The loss function is

$$\mathcal{L}_{\text{proto}}^{u} = \frac{1}{|M_u|} \sum_{(f_u, \hat{y_u}) \in M_u} \mathcal{L}_{\text{CE}} \left(\frac{\text{sim}\left(f^{\text{ul}}, c_k^l\right)}{\tau_t}, \hat{y_u} \right). \tag{9}$$

This loss constrains the distribution of unlabeled features using the strong semantic guidance provided by labeled prototypes, thereby preventing unlabeled prototypes from drifting away from the true semantic space due to pseudo-label biases (e.g., overgeneration of majority-class pseudo-labels).

Through these bidirectional cross-constraints, the model not only leverages the reliable supervision from labeled data to suppress prototype drift in the unlabeled branch but also exploits the rich information contained in unlabeled data to compensate for the scarcity of minority-class samples in the labeled set. Ultimately, this mechanism promotes a more balanced distribution of majority and minority classes in the feature space, effectively mitigating the feature learning bias induced by class imbalance in RSI.

E. Loss Function

The SSSS framework proposed in this article adopts a multicomponent loss function. Model training is realized by fusing supervised loss, unsupervised consistency loss, and prototype contrastive loss. The total loss function is defined as

$$\mathcal{L} = \mathcal{L}_{\text{sup}} + \alpha \cdot \mathcal{L}_{\text{unsup}} + \beta \cdot \mathcal{L}_{\text{proto}}^{l} + \gamma \cdot \mathcal{L}_{\text{proto}}^{u} \tag{10}$$

where $\mathcal{L}_{\text{sup}}$ is the supervised loss for labeled data, $\mathcal{L}_{\text{unsup}}$ is the consistency loss for unlabeled data, and $\mathcal{L}_{\text{proto}}^{l}$ and $\mathcal{L}_{\text{proto}}^{u}$ are the prototype contrastive losses based on labeled and unlabeled data, respectively. α is the weight of the unsupervised loss, β and γ are the weight coefficients of the prototype losses, with $\beta + \gamma = 1$.

The supervised loss $\mathcal{L}_{\text{sup}}$ uses cross-entropy loss to calculate the classification loss of labeled data

$$\mathcal{L}_{\text{sup}} = \text{CE}\left(p^l, y^l\right) \tag{11}$$

where p^l is the model's prediction for labeled samples, and y^l is the real label.

The unsupervised consistency loss constrains the learning of unlabeled data through data augmentation consistency and uses an entropy threshold to filter high-confidence pseudo-labels

$$\mathcal{L}_{\text{unsup}} = \mathbb{E}\left[\text{CE}\left(p^u, y^u\right) \cdot \mathcal{I}\left(H\left(p^u\right) < \tau_e\right)\right] \tag{12}$$

where p^u is the high-confidence pseudo-label of weakly augmented samples (entropy is calculated by $H(\cdot)$, and $\tau_e$is the entropy threshold), and y^u is the prediction of strongly augmented samples.

IV. Experimental Results

A. Datasets

The ISPRS Potsdam dataset consists of 38 aerial images, each containing four spectral channels (near infrared (NIR), red, green, and blue) with a spatial resolution of 5 cm and an image size of 6000 × 6000 pixels. Six land-cover classes are annotated: impervious surfaces, buildings, low vegetation (LV), trees, cars, and background. Following the dataset split proposed in [8], 18 images are used for training, 6 for validation, and 14 for testing. To reflect real-world application scenarios and improve the utilization of data and computational resources, a two-stage cropping strategy is applied to the training set. First, each training image is nonoverlappingly divided into four 1500 × 1500 subimages. A subset of these subimages is then randomly selected as labeled data at three ratios (1/18, 1/36, and 1/72), while the remaining ones serve as unlabeled data to maintain the relative completeness of different land-cover types. Specifically, **when the ratio is 1/72, only one** 3000 × 3000 **subimage is selected as labeled data.** Second, throughout model training, a 512 × 512 sliding window is used to randomly crop these subimages to generate the final training samples. This strategy ensures compatibility with computational constraints while improving data utilization efficiency.

The ISPRS Vaihingen dataset is another high-resolution aerial imagery dataset, consisting of 33 images of varying sizes with a spatial resolution of 9 cm. It contains three spectral bands—NIR, red (R), and green (G)—and the annotated land-cover categories are consistent with those of the ISPRS Potsdam dataset. For dataset partitioning, three images (IDs 11, 28, and 34) are used for validation, two images (IDs 15 and 30) for testing, and the remaining images for training. The same two-stage cropping strategy is applied. In the first stage, each training image is nonoverlappingly divided into four subimages, from which a subset is randomly selected as labeled data at the same ratios (1/18, 1/36, and 1/72), while the rest serve as unlabeled data to preserve land-cover completeness. In the second stage, during training, 512 × 512 random crops are generated from these subimages to produce the final training samples, further improving computational efficiency and data utilization.

B. Implementation Details

To ensure experimental fairness, all experimental parameters are kept identical across trials. The model is trained for 50 epochs, with a batch size of 8 set for both labeled and

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TABLE I
ABLATION EXPERIMENTS ON POTSDAM DATASET

Base	$\mathcal{L}^{l}_{proto}$	$\mathcal{L}^{u}_{proto}$	1/18		1/36		1/72	
			mIou	mF1	mIou	mF1	mIou	mF1
✓			78.38	87.68	75.48	85.77	45.48	61.84
✓	✓		78.72	87.91	75.86	86.06	57.93	72.51
✓		✓	78.57	87.86	75.93	86.19	55.49	70.27
✓	✓	✓	**79.02**	**88.09**	**77.00**	**86.82**	**60.46**	**75.08**

TABLE II
ABLATION EXPERIMENTS ON VAIHINGEN DATASET

Base	$\mathcal{L}^{l}_{proto}$	$\mathcal{L}^{u}_{proto}$	1/18		1/36		1/72	
			mIou	mF1	mIou	mF1	mIou	mF1
✓			69.87	81.71	65.79	78.56	54.74	69.74
✓	✓		70.80	82.31	67.98	80.38	56.17	70.86
✓		✓	70.87	82.42	67.84	80.13	56.66	71.49
✓	✓	✓	**71.80**	**83.11**	**68.49**	**80.74**	**57.64**	**72.83**

TABLE III
INDEPENDENT ABLATION OF THE DYNAMIC CW MECHANISM ON POTSDAM (1/18 RATIO)

Strategy	IS	Building	LV	Tree	Car	mIoU
Baseline	82.91	89.90	72.57	68.14	78.38	78.38
w/o CW	82.70	89.69	72.13	69.16	78.33	78.40
w/ CW	**83.34**	**90.67**	**78.35**	**79.02**	**83.34**	**79.02**

unlabeled data, and a pseudo-label confidence threshold of 0.5. For the clustering-based learning approach, a stochastic gradient descent (SGD) optimizer is adopted. Specifically, the base learning rate is set to 0.01 for the encoder and 0.001 for the decoder; after each iteration, the initial learning rate is updated using the formula

$$lr_{base} \times (1 - (iter/total_{iter}))^{0.9}. \tag{13}$$

Additionally, the momentum factor and weight decay coefficient are configured as 0.9 and 0.0001, respectively. The tail-class enhancement factor δ in (6) is empirically set to 3.5 to prioritize the learning of rare categories and mitigate the gradient dominance of majority classes. The network has been implemented using the PyTorch framework and is trained on a GeForce RTX 3090 GPU with 24 GB of memory.

During the inference phase, a sliding window cropping strategy is employed to extract image patches from the original test images, which are then processed sequentially. Three evaluation metrics are used to assess the model's performance: mean intersection over union (mIoU), $F1$-score, and overall accuracy (OA). The code can be accessed at the following URL: https://github.com/LiZhangwhu/CPBCF

$$OA = \frac{TP + TN}{TP + TN + FP + FN} \tag{14}$$

$$IOU = \frac{TP}{TP + FP + FN}, \quad mIOU = \frac{\sum_{i=1}^{n} IOU_i}{n} \tag{15}$$

$$precision = \frac{TP}{TP + FP}, \quad recall = \frac{TP}{TP + FN} \tag{16}$$

$$F1 = 2 \times \frac{precision \times recall}{precision + recall}, \quad mF1 = \frac{\sum_{i=1}^{n} F_{1i}}{n}. \tag{17}$$

TP is the number of correctly predicted positive samples, TN is the number of correctly predicted negative samples, FP is the number of incorrectly predicted positive samples, and FN is the number of incorrectly predicted negative samples.

C. Ablation Study

To comprehensively assess the effectiveness of the proposed framework, WSCL [50] is employed as the baseline model. Ablation studies are performed on the ISPRS Potsdam and Vaihingen datasets by incrementally introducing the proposed prototypical loss components. The results are reported in Table I (Potsdam) and Table II (Vaihingen).

1) Effect of $\mathcal{L}^{l}_{Proto}$: The goal of $\mathcal{L}^{l}_{\text{proto}}$ is to alleviate pseudo-label bias amplification and prototype drift caused by class imbalance in SSSS. Under scarce annotation, pseudo-labels tend to be dominated by majority classes, causing unlabeled features to gradually collapse toward majority-class prototypes and further weakening the representation of rare classes. By introducing $\mathcal{L}^{l}_{\text{proto}}$, unstable unlabeled prototypes are realigned through constraints from reliable labeled prototypes, thereby suppressing majority-class bias and stabilizing the semantic distribution. Through experimental observations, consistent performance gains are observed across all label-split settings, with the most pronounced improvements in the extremely low-label 1/72 scenario. On the Potsdam dataset, mIoU increases from 45.48% to 57.93% (+ 12.45%), and on the Vaihingen dataset, mIoU improves from 54.74% to 56.17%. These results demonstrate that $\mathcal{L}^{l}_{\text{proto}}$ effectively corrects pseudo-label deviations and provides a reliable anchor for maintaining a stable unlabeled feature space.

2) Effect of $\mathcal{L}^{u}_{Proto}$: $\mathcal{L}^{u}_{\text{proto}}$ addresses the insufficient feature representation of rare classes, which arises from extremely limited labeled samples in RSI. Labeled prototypes of rare classes are easily influenced by noise and thus fail to provide stable guidance. To mitigate this issue, $\mathcal{L}^{u}_{\text{proto}}$ enhances rare-class prototypes by leveraging pseudo-label-based prototypes computed from abundant unlabeled samples, thereby enriching feature structure and improving discriminability. The experimental results validate this mechanism. On the Vaihingen dataset, mIoU increases from 65.79% to 67.84% under the 1/36 setting, and from 54.74% to 56.66% under the 1/72 setting. Similar trends are observed on Potsdam, where rare-class mIoU improves by an average of 3.21% under the 1/72 split. Unlike $\mathcal{L}^{l}_{\text{proto}}$, which mainly suppresses bias from dominant classes, $\mathcal{L}^{u}_{\text{proto}}$ directly strengthens the representation of rare classes. This complementary role enables the model to maintain balanced learning even under extreme annotation scarcity.

3) Effect of the Dynamic CW Mechanism: Table III isolates the impact of the dynamic CW mechanism on the Potsdam dataset (1/18). The results show that prototype constraints alone (w/o CW) yield a negligible mIoU gain of 0.02%, with the minority "Car" class even suffering a slight decline (78.38% to 78.33%). This confirms CW as an indispensable enabling foundation. In imbalanced remote sensing scenes, majority-class dominance causes "prototype drift" or

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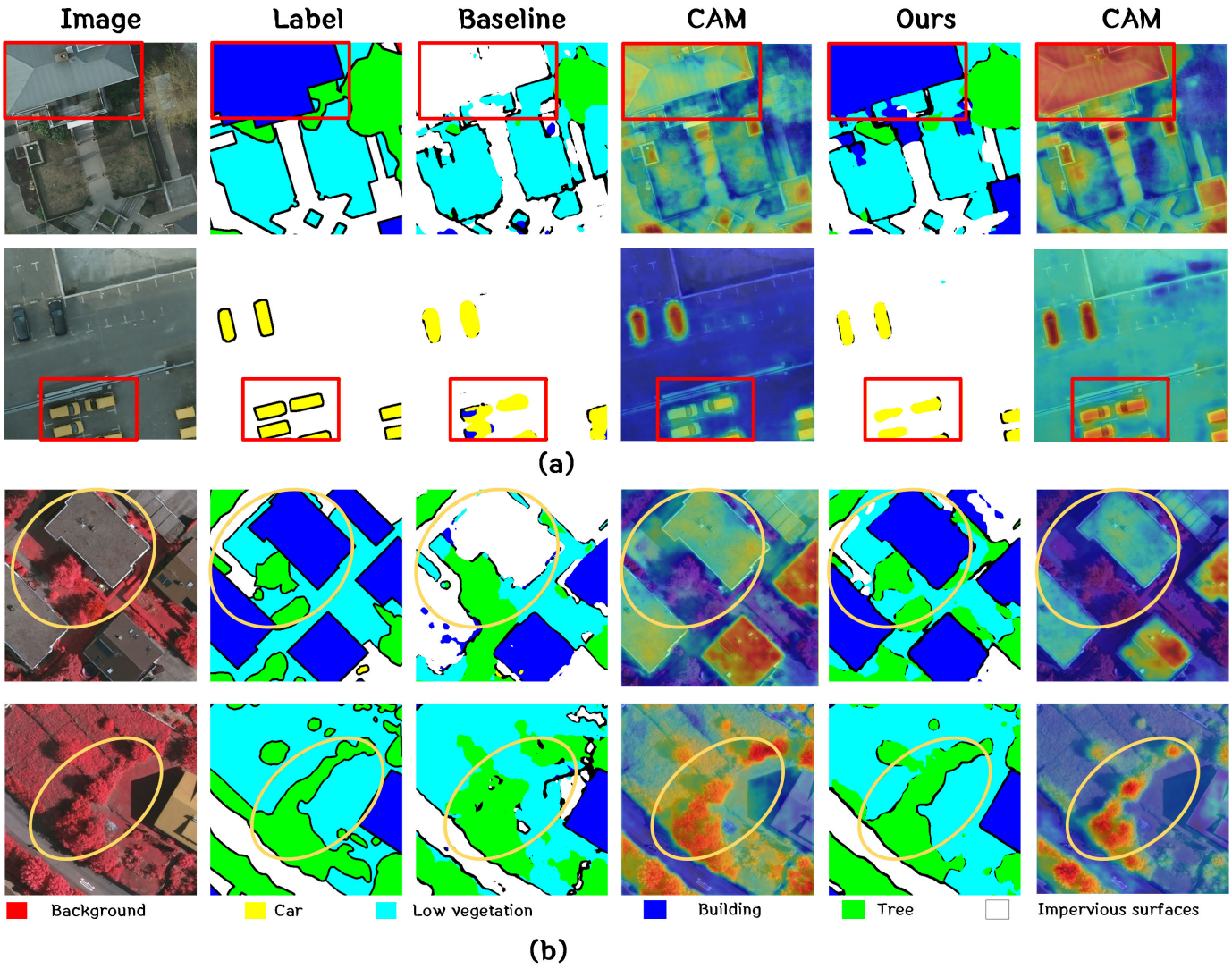


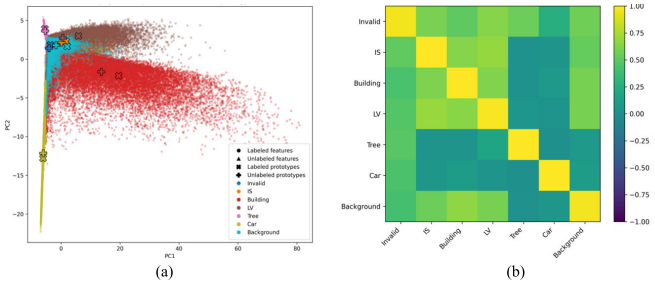
Fig. 3. Comparison of prediction results between Baseline and our method. We use a class activation map (i.e., CAM) [49] to perform a simple feature visualization on the fourth and sixth columns. Among them, the first and third rows show the CAMs of "Building," the second row corresponds to "Car," and the fourth row represents "Tree." (a) Potsdam. (b) Vaihingen.

semantic "drowning." The CW mechanism recalibrates learning intensity, allowing rare categories to establish stable semantic anchors early in training. Consequently, enabling CW leads to significant jumps in Tree (+9.86%) and Car (+5.01%) IoU, proving that bidirectional alignment requires semantically stable targets provided by the CW mechanism to function effectively.

4) Effect of Combining $\mathcal{L}^{l}_{Proto}$ and $\mathcal{L}^{u}_{Proto}$: Combining both loss terms yields the best overall performance, demonstrating strong complementarity. $\mathcal{L}^{l}_{\text{proto}}$ suppresses majority-class pseudo-label bias, whereas $\mathcal{L}^{u}_{\text{proto}}$ enhances rare-class feature representation. Together, they form a bidirectional prototype constraint mechanism that dynamically balances prototype learning. This design ensures that unlabeled data contribute informative structural cues without introducing semantic drift. The improvement is substantial. On the Potsdam dataset, the combined model achieves the highest performance across all settings, reaching 79.02% mIoU and 88.09% mF1 under the 1/18 split. Under the extremely sparse 1/72 setting, mIoU increases from 45.48% to 60.46%. Consistent gains are observed on the Vaihingen dataset. Qualitative results in Fig. 3 further illustrate reduced confusion between semantically similar categories, such as "Building" vs. "Impervious Surface" and "Tree" versus "LV," as well as improved segmentation of small objects like "Car." These findings confirm that the proposed bidirectional prototype constraints are both theoretically sound and practically effective, significantly enhancing distribution stability and robustness in semi-supervised training. To further investigate the stability of the proposed CPBCF, we visualize the prototype alignment and feature distribution in Fig. 4. The diagonal-dominant heatmap [Fig. 4(a)] and the close clustering of labeled/unlabeled prototypes in the PCA projection [Fig. 4(b)] demonstrate that our framework effectively prevents prototype drift even under sparse annotations.

5) Effect of Labeled Data Ratio: To further analyze the influence of labeled data ratio on segmentation performance

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(a) (b)

Fig. 4. Prototype consistency and feature distribution analysis.

and evaluate the adaptability of the proposed method, we compare the results under three label ratios (1/18, 1/36, and 1/72) in Tables I and II. Overall, segmentation performance exhibits a strong positive correlation with the labeled data ratio, although the degree of performance degradation varies across datasets. When the ratio decreases from 1/18 to 1/72, the baseline mIoU on the Potsdam dataset drops by more than 40%, whereas the reduction on Vaihingen is about 22%. This discrepancy is primarily attributed to dataset characteristics: Potsdam contains more complex and fragmented land-cover structures, making the model more dependent on supervised signals. This observation aligns with the visual results in Fig. 5, where the baseline exhibits severe class confusion under Potsdam 1/72—such as blurred boundaries between "Building" and "Vegetation"—but produces more accurate results under the 1/18 setting. The proposed method effectively mitigates the adverse effects of reduced supervision. On Potsdam 1/72, the proposed CPBCF achieves a significant performance leap, improving mIoU and mF1 by 15.00% and 13.24%, respectively, compared with the baseline. This substantial enhancement is directly attributed to the model's ability to rectify the severe distribution discrepancy inherent in sparse sampling. As quantified in Fig. 1(b) and (c), the 1/72 random partition in Potsdam triggers a "distributional drift" where historically dominant categories. By aligning labeled prototypes with the more representative unlabeled feature space, our framework effectively compensates for the biased priors learned from these distorted label distributions. In contrast, the performance gain on Vaihingen (2.90% for mIoU) is more moderate because its labeled and unlabeled subsets exhibit more consistent, i.i.d.-like distributions. The consistent improvements across both datasets, despite their varying degrees of distribution shift, further confirm the robustness and strong adaptability of CPBCF in real-world, nonidealized semi-supervised scenarios.

6) Generalization to Multibranch Frameworks: To evaluate the generalization capability of the proposed method, we integrate it into PS-MT [40], a representative multibranch SSSS framework. Unlike the single-branch architecture used in the main experiments, multibranch frameworks strengthen consistency regularization by introducing multiple prediction heads; however, they still suffer from pseudo-label bias under extreme annotation scarcity. Experimental results on Potsdam (Table IV) and Vaihingen (Table V) show consistent performance improvements across all label ratios (1/18, 1/36, and 1/72). For example, on Potsdam 1/18, the proposed method improves mIoU from 80.18% to 82.40% and OA from 87.75% to 90.14%. Under the extremely low-label 1/72 setting, the improvements become even more significant: mIoU increases from 19.36% to 34.45%, mF1 from 31.77% to 48.20%, and OA rises by more than 20%. Similar trends are observed on the Vaihingen dataset, where mIoU and mF1 under the 1/18 split increase by 13.65% and 12.48%, respectively. These results verify that the proposed prototype constraints effectively suppress pseudo-label bias and enhance rare-class discriminability. The consistent gains across different architectures demonstrate the strong transferability and robustness of the method in diverse semi-supervised segmentation frameworks.

7) Sensitivity Analysis of Hyperparameters: To evaluate the influence of the relative weights assigned to the two prototype-level constraints, we conduct a sensitivity analysis of the hyperparameters β and γ using the 1/36 label ratio as an example. Here, β controls the strength of the constraint from labeled prototypes to unlabeled prototypes, whereas γ regulates the reverse constraint. As shown in Table VI, the model achieves

TABLE IV
EXPERIMENTS ON DIFFERENT BASELINE WITH POTSDAM DATASET

Ratio	Model	Evaluation		
		mlou	mF1	OA
1/18	PS-MT	80.18	88.81	87.75
	PS-MT-ours	82.40	90.18	90.14
1/36	PS-MT	74.92	85.53	84.86
	PS-MT-ours	80.59	89.01	88.70
1/72	PS-MT	19.36	31.77	39.54
	PS-MT-ours	34.45	48.20	60.51

TABLE V
EXPERIMENTS ON DIFFERENT BASELINE WITH VAIHINGEN DATASET

Ratio	Model	Evaluation		
		mlou	mF1	OA
1/18	PS-MT	42.58	56.14	64.09
	PS-MT-ours	56.23	68.62	77.79
1/36	PS-MT	31.71	42.94	53.87
	PS-MT-ours	34.92	48.85	55.60
1/72	PS-MT	25.41	37.19	47.33
	PS-MT-ours	28.92	42.53	49.27

TABLE VI
EFFECT OF THE HYPERPARAMETERS ON THE POTSDAM DATASET

β	γ	Evaluation		
		mIou	mF1	OA
0.1	0.9	76.22	86.30	85.24
0.3	0.7	**77.00**	**86.82**	**85.77**
0.5	0.5	76.49	86.53	85.46
0.7	0.3	76.36	86.39	85.23
0.9	0.1	75.97	86.12	85.14

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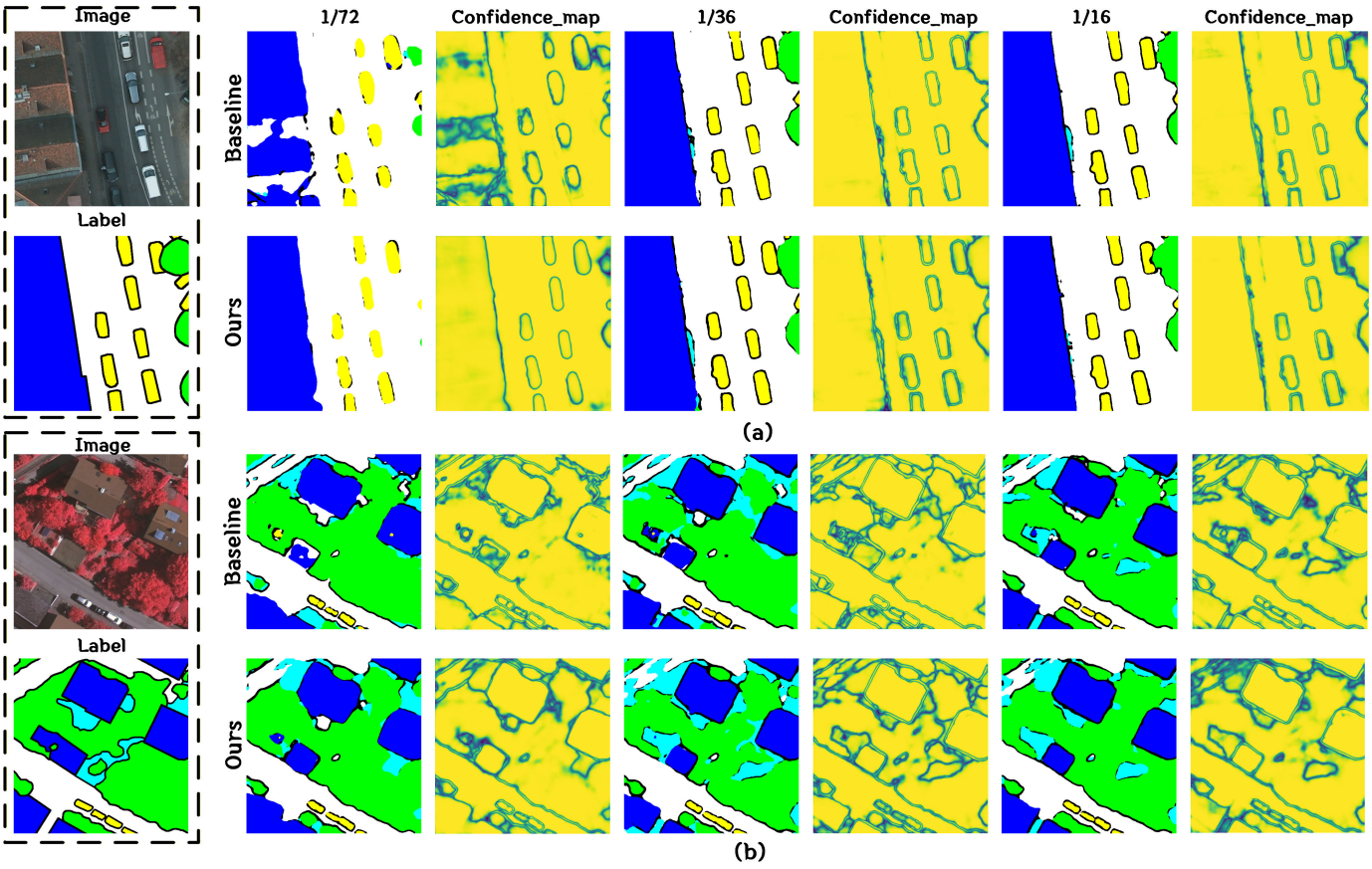


Fig. 5. Prediction results and their corresponding confidence map with different supervised ratios of training data. (a) Potsdam. (b) Vaihingen.

its highest performance when $\beta = 0.3$ and $\gamma = 0.7$. When β is too small, the supervision from labeled prototypes becomes insufficient, causing unlabeled prototypes to drift in regions influenced by noisy pseudo-labels. Conversely, an excessively large β (e.g., 0.7) overemphasizes labeled prototypes and suppresses the beneficial structural cues contained in unlabeled data, thereby limiting improvements. A similar pattern is observed for γ. An overly large γ introduces noise due to excessive reverse pulling from uncertain unlabeled prototypes, while a too-small γ weakens the ability to exploit complementary information from unlabeled samples—particularly for refining class boundaries and enhancing tail-class representations. Overall, the combination $\beta = 0.3$ and $\gamma = 0.7$ provides the best balance between stability and adaptability. This setting enables the bidirectional prototype constraints to simultaneously correct pseudo-label bias and enrich prototype representations with large-scale unlabeled data, leading to the optimal segmentation performance.

8) Computational Complexity and Practical Utility: To further evaluate the practical utility of the proposed framework, we conduct a comprehensive analysis of computational complexity. All metrics are measured with a batch size of 8 on 512×512 images across seven semantic classes. As summarized in Table VII, our CPBC module achieves a superior balance between performance enhancement and computational efficiency. In terms of inference, the CPBC-enhanced models maintain identical parameter counts and inference FLOPs compared to their respective baselines (WSCL and PS-MT). For instance, both WSCL and CPBC achieve an inference speed of 58.4 FPS. This zero-inference overhead is attributed to the architectural design where CPBC serves as a training-only auxiliary branch. During the inference phase, the prototype-based consistency regularization is completely pruned, leaving the core segmentation backbone and decoder unchanged for deployment. Regarding training consumption, the introduction of CPBC leads to a marginal increase in training FLOPs. This negligible training cost stems from the efficient implementation of prototype generation. Unlike the heavy $O(H \times W)$ pixel-level convolutions in the backbone, CPBC derives class-wise prototypes via global average pooling (GAP) and enforces consistency through lightweight vector-level distance metrics. These operations

TABLE VII
COMPUTATIONAL COMPLEXITY OF DIFFERENT MODEL

model	Parameters/M	Inference FLOPs/G	Training FLOPs/T	FPS
WSCL	40.35	366.21	17.578	58.4
+CPBC	40.35	366.21	**17.598**	58.4
PS-MT	125.29	306.14	9.797	55.42
+CPBC	125.29	306.14	**9.821**	55.42

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TABLE VIII
COMPARATIVE EXPERIMENT WITH POTSDAM DATASET

Labeled ratio	Model	Iou/F1					mIou/mF1	OA
		IS	BD	LV	Tree	Car		
Supervised	DeepLabv3+	86.70/92.88	92.86/96.30	76.01/86.37	77.61/87.40	79.85/88.79	82.60/90.35	90.06
1/18	ST	82.53/90.43	89.42/94.42	70.79/82.90	70.36/82.6	69.61/82.08	76.54/86.48	87.77
	Fixmatch	81.61/89.87	89.77/94.61	70.46/82.67	70.08/82.41	72.75/84.22	76.93/86.76	87.59
	DWL	80.27/89.05	87.71/93.46	69.71/82.15	71.23/83.20	67.17/80.36	75.22/85.64	86.78
	PS-MT	81.3/89.69	87.89/93.55	70.49/82.69	73.07/84.44	**88.16/93.71**	80.18/88.81	87.75
	BCNet	81.12/89.58	87.87/93.55	68.28/81.15	67.34/80.49	88.01/93.62	78.53/87.68	86.62
	WSCL	82.91/90.65	89.90/94.68	72.57/84.11	68.14/81.05	78.38/87.88	78.38/87.68	86.78
	RS-MUCA	71.63/83.47	75.99/86.36	69.29/81.86	66.98/80.22	82.70/90.53	73.32/84.49	86.87
	PS-MT-ours	**85.65/92.27**	**91.20/95.40**	**73.46/84.70**	**74.59/85.45**	87.09/93.10	**82.40/90.18**	**90.14**
	WSCL-ours	83.34/90.91	90.67/95.11	72.51/84.06	70.26/82.53	78.35/87.86	79.02/88.09	87.28
1/36	ST	77.06/87.07	87.70/93.35	65.83/79.39	66.47/79.86	67.97/80.93	73.01/84.13	84.79
	Fixmatch	77.10/87.05	85.11/92.40	67.74/80.77	67.13/80.33	70.99/83.04	73.73/84.70	84.70
	DWL	78.17/87.75	85.32/92.08	67.40/80.52	65.85/79.41	70.01/82.36	73.35/84.42	85.01
	PS-MT	78.19/87.76	82.13/90.19	66.67/80.00	69.11/81.73	78.51/87.96	74.92/85.53	84.86
	BCNet	79.97/88.87	85.54/92.21	67.19/80.38	68.01/80.96	87.55/93.36	77.65/87.16	86.05
	WSCL	**80.17/88.99**	86.95/93.02	69.25/81.83	63.05/77.34	78.00/87.64	75.48/85.77	84.84
	RS-MUCA	71.19/83.17	73.48/84.71	66.02/79.54	58.81/74.07	83.62/91.08	70.63/82.51	85.47
	PS-MT-ours	84.26/91.46	89.40/94.40	69.49/82.00	**71.54/83.41**	**88.25/93.76**	**80.59/89.01**	**88.70**
	WSCL-ours	79.56/88.62	85.89/92.41	**69.94/82.31**	68.76/81.49	78.30/87.83	76.49/86.53	85.46
1/72	ST	47.04/63.99	45.11/62.17	51.46/67.95	**55.42/71.32**	47.25/64.18	49.26/65.92	65.60
	Fixmatch	50.24/66.88	35.08/51.94	60.98/75.76	62.35/76.81	58.54/73.85	53.44/69.05	67.38
	DWL	41.58/58.74	45.03/62.10	48.24/65.09	59.98/74.98	51.05/67.59	49.18/65.70	63.44
	PS-MT	21.53/35.44	20.50/34.02	30.97/47.30	15.83/27.33	7.97/14.76	19.36/31.77	39.54
	BCNet	24.67/39.08	38.86/51.70	34.67/51.49	49.62/66.33	37.24/54.27	36.21/52.67	50.33
	WSCL	56.33/72.56	27.11/42.66	64.51/78.53	51.23/60.05	**64.68/78.55**	53.83/67.94	67.15
	RS-MUCA	42.22/59.37	18.80/31.64	41.49/58.64	37.46/54.50	3.18/6.16	28.63/42.06	68.73
	PS-MT-ours	47.32/64.24	30.30/46.51	49.85/64.53	45.91/68.01	1.86/3.65	32.44/48.20	60.51
	WSCL-ours	**65.03/78.81**	**67.58/80.64**	**67.15/80.35**	51.05/67.59	51.53/68.01	**60.46/75.08**	**77.34**

account for a negligible portion of the total computational graph. These results demonstrate that CPBC is a highly efficient "plug-and-play" component, making it particularly suitable for large-scale remote sensing tasks where computational resources for deployment are often constrained.

D. Comparison With State-of-the-Art Methods

To comprehensively evaluate the effectiveness of the proposed bidirectional prototype constraint mechanism, we compare it with several state-of-the-art SSSS methods (ST [51], FixMatch [43], DWL [52], PS-MT [40], BCNet [8], WSCL [50], and a transformer-based method RS-MUCA [53]) on the ISPRS Potsdam and Vaihingen datasets (as shown in Tables VIII and IX). For reference, DeepLabV3 + under full supervision is included to quantify the performance gap between semi-supervised and fully supervised settings. To further validate the generalization capability, our module is additionally integrated into the multibranch framework PS-MT (denoted as PS-MT-ours). All experiments are conducted under three label ratios (1/18, 1/36, and 1/72), covering scenarios from relatively sufficient labels to extremely scarce supervision.

Under the 1/18 setting, PS-MT-ours achieves 90.14% OA on Potsdam—surpassing even the fully supervised DeepLabV3 + model. This result highlights the ability of our method to effectively mine reliable semantic information from unlabeled data and construct a more stable class feature space through prototype-level constraints, thereby reducing dependence on large volumes of annotated samples. In the moderately scarce 1/36 setting, the proposed approach attains the best per-

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TABLE IX
COMPARATIVE EXPERIMENT WITH VAIHINGEN DATASET

Labeled ratio	Model	Iou/F1					mIou/mF1	OA
		IS	BD	LV	Tree	Car		
Supervised	DeepLabv3+	80.64/89.28	90.56/95.05	66.67/80.00	78.90/88.20	66.23/79.68	76.60/86.44	86.52
1/18	ST	76.49/86.54	86.09/92.52	60.58/75.45	71.75/83.55	50.70/67.28	69.12/81.10	**84.30**
	Fixmatch	74.74/85.68	85.16/91.99	60.48/75.38	71.99/83.72	48.31/65.14	68.14/80.35	83.92
	DWL	76.29/86.55	85.56/92.22	60.66/75.51	70.29/82.55	50.80/67.37	68.72/80.84	83.78
	PS-MT	46.24/63.24	62.67/77.05	6.49/12.19	64.08/78.11	33.43/50.10	42.58/56.14	64.09
	BCNet	56.73/72.39	78.86/88.18	31.01/47.34	71.86/83.63	**62.76/77.12**	60.24/73.73	74.92
	WSCL	75.33/85.93	87.74/93.58	57.15/72.73	**75.09/85.78**	58.83/74.08	70.83/82.42	81.72
	RS-MUCA	75.36/85.95	79.73/88.72	60.33/75.26	69.12/81.74	48.6/65.41	66.63/79.41	81.64
	PS-MT-ours	74.04/85.09	73.24/84.55	51.30/67.81	68.86/81.56	13.68/24.07	56.23/68.62	77.79
	WSCL-ours	**77.31/87.20**	**87.80/93.58**	**61.33/76.03**	74.64/85.48	57.93/73.36	**71.80/83.11**	83.11
1/36	ST	68.38/81.22	78.08/87.69	51.06/67.60	58.60/73.90	46.70/63.67	60.56/74.82	77.23
	Fixmatch	73.21/84.53	83.12/90.78	47.98/64.85	50.38/67.00	42.37/59.52	59.41/73.34	76.50
	DWL	70.17/82.47	79.96/88.86	51.36/67.86	54.95/70.92	36.70/53.69	58.63/72.76	76.88
	PS-MT	34.98/51.83	61.12/75.87	1.10/2.10	52.46/68.82	9.86/17.96	31.71/42.94	53.87
	BCNet	46.97/63.92	63.66/77.29	21.83/35.84	55.32/71.23	39.16/56.28	45.39/61.01	63.71
	WSCL	**73.56/84.76**	**85.84/92.38**	55.72/71.55	**71.74/83.54**	**55.60/71.46**	68.49/80.84	80.54
	RS-MUCA	72.99/84.39	77.43/87.28	57.64/73.13	69.74/82.17	15.97/27.55	58.75/70.90	79.01
	PS-MT-ours	57.35/72.89	46.62/63.59	22.32/36.07	41.65/58.81	6.64/12.46	34.92/48.85	55.60
	WSCL-ours	72.60/84.12	84.22/91.43	**56.91/72.54**	64.63/78.52	50.34/66.97	**65.74/78.72**	**85.22**
1/72	ST	53.88/70.03	56.82/72.47	39.20/56.32	53.90/70.04	30.78/47.07	46.92/63.19	66.98
	Fixmatch	**65.03/78.81**	67.91/80.89	45.37/62.42	36.42/53.40	27.36/42.96	48.42/63.69	68.61
	DWL	57.71/73.18	56.81/72.46	45.14/62.20	63.81/77.91	23.83/38.49	49.46/64.85	71.27
	PS-MT	35.98/52.92	14.27/24.98	3.01/5.22	54.29/70.37	19.47/32.60	25.41/37.19	47.33
	BCNet	45.95/62.96	58.42/73.76	23.28/37.77	56.24/71.99	27.54/43.19	42.29/57.93	61.97
	WSCL	62.58/76.99	**68.31/81.17**	35.78/52.70	**66.70/80.02**	49.92/66.60	56.66/71.49	72.04
	RS-MUCA	61.89/76.46	67.91/80.89	16.63/28.52	59.29/74.44	**55.03/70.99**	52.15/66.26	66.31
	PS-MT-ours	39.81/56.95	35.35/52.23	19.55/32.70	45.68/62.71	4.19/8.05	28.92/42.53	49.27
	WSCL-ours	61.98/76.53	66.20/79.67	**46.88/63.84**	62.82/77.16	50.31/66.94	**57.64/72.83**	**73.23**

formance on both datasets and consistently outperforms all competing methods. Remarkably, without additional data augmentation, the IoU of the rare "Car" class improves by 9.8%, demonstrating that the proposed prototype constraints effectively mitigate weak feature representation and unstable prototype formation for rare classes under limited supervision.

We visualized the segmentation results under the extremely low-label 1/72 setting (as shown in Fig. 6). Under the 1/72 setting, where pseudo-label bias and class imbalance are most pronounced, mainstream semi-supervised methods exhibit severe performance degradation (e.g., ST and FixMatch obtain only 47.04% and 50.24% mIoU, respectively). In contrast, our method achieves 65.03%/78.81% (mIoU/mF1) and 77.34% OA on Potsdam, outperforming all competitors by nearly 9% mIoU. These results validate that the proposed bidirectional prototype constraints effectively stabilize the learning process: L^{l}_{proto} corrects pseudo-label bias and suppresses majority-class drift, while L^{u}_{proto} enhances rare-class representation by leveraging large volumes of unlabeled data. Furthermore, PS-MT-ours exhibits substantial improvement, which from 30.30%/46.51% to 47.32%/64.24%, further confirming the strong generalization and robustness of our method across different semi-supervised frameworks.

It is worth noting that under the 1/72 split, most rare classes contain almost no labeled samples, causing the model to rely heavily on early pseudo-label frequency signals. As a result, the "LV" class—having relatively high pixel prevalence in unlabeled data—is frequently predicted with high confidence as a majority class. Due to its stable pseudo-label distribution and reinforced prototype updates, the IoU differences across methods on LV become smaller and generally higher than those of other classes. This observation underscores the necessity of the proposed method: its core advantage lies not in further improving pseudo-majority classes (e.g., LV), but in

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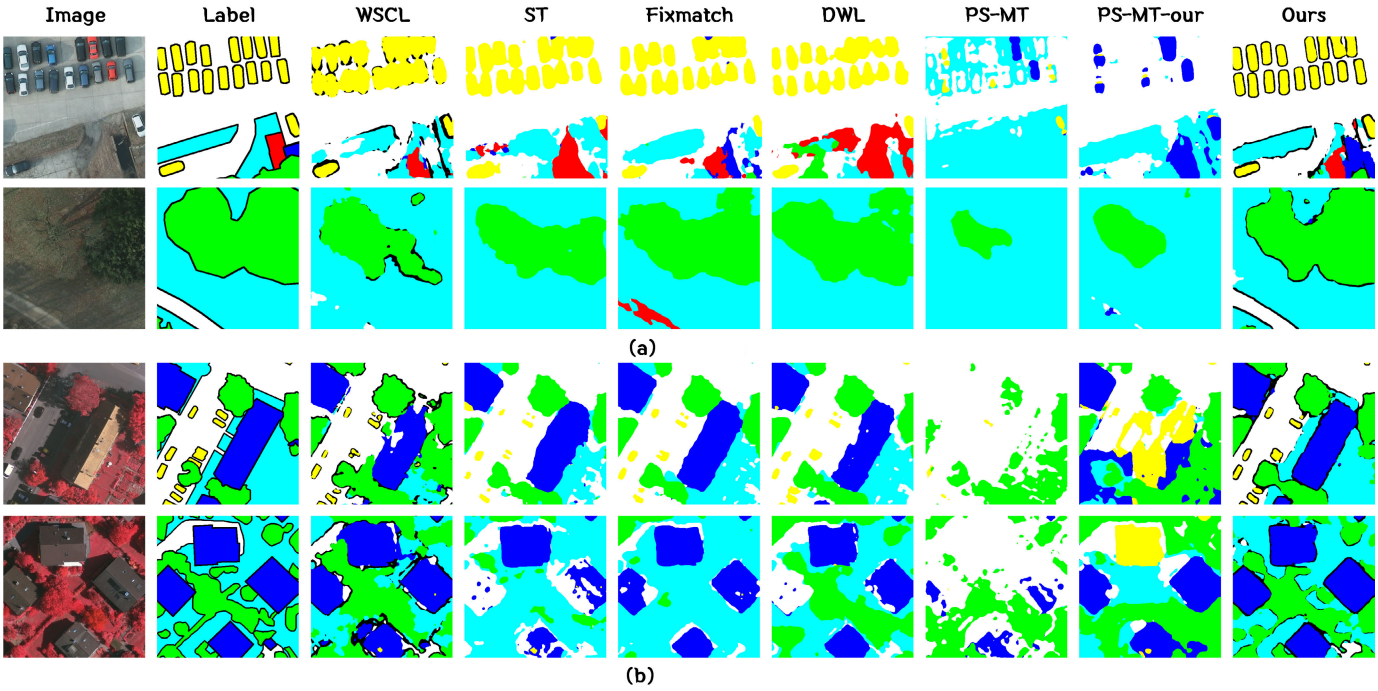


Fig. 6. Comparison of prediction results between different methods. (a) Potsdam. (b) Vaihingen.

mitigating severe bias against true minority categories (e.g., Tree and Clutter) during pseudo-label generation, ensuring reliable prototype representation and balanced performance under extremely weak supervision. Furthermore, it is critical to address an atypical performance degradation observed in the PS-MT-ours variant for the "Car" class under the extremely sparse 1/72 setting on the Potsdam dataset, where the IoU drops to 1.86%. A detailed statistical analysis reveals the underlying cause: in this specific 1/72 labeled subset, the "Car" class is characterized by both its extremely small spatial scale and severe sample scarcity. Under such extreme sparsity, the labeled prototype (c_k^l) for this minority class is highly susceptible to noise interference or semantic collapse. Consequently, during the bidirectional alignment process within a highly coupled multibranch framework, the unlabeled features are inadvertently misdirected toward an erroneous semantic space, triggering negative transfer. In stark contrast, under the 1/36 and 1/18 settings, where the annotated pixels for the "Car" class are sufficiently represented, the model maintains a normal and highly competitive IoU (e.g., an improvement of 9.8% under the 1/36 setting). This specific phenomenon explicitly reveals the behavioral boundaries of prototype-based constraints under near-zero-shot conditions.

V. Conclusion

This article presents a CPBCF to alleviate class imbalance and pseudo-label bias in SSSS of RSI. By jointly stabilizing labeled and unlabeled prototypes, the proposed method effectively enhances the representation consistency of minority classes and strengthens pseudo-label reliability under limited supervision. Experiments on Potsdam and Vaihingen demonstrate clear advantages over mainstream semi-supervised methods across different label ratios, and the framework also shows strong generalizability when embedded into multibranch architectures.

Although effective, the framework's performance inherently relies on stable prototype estimation, thereby explicitly defining its applicability boundaries. Specifically, the labeled dataset must contain a minimum viable number of samples for each category to support basic semantic representations. For "near-zero-shot" scenarios where a specific class is completely absent or suffers from extreme sample scarcity (as observed in the aforementioned "Car" class under the 1/72 ratio), a pure prototype alignment mechanism may face the risk of failure due to prototype collapse. Future work will explore integrating adaptive noise filtering and pseudo-label debiasing techniques to further enhance the model's reliability and prevent negative transfer under extreme class imbalance and severely limited sample conditions. Furthermore, extensions to multimodal and large-scale foundation models will be investigated to construct more robust initial prototypes.

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